

Midterm 2 Review Worksheet

1. Are each of the following statements true or false?

- (a) If W is a subspace of V then every linear combination of vectors from V is also in W .
- (b) If $B = (x + 1, x - 4)$ is an ordered basis for P_1 , and $C_B(p) = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$, then $p = 7x - 2$.
- (c) The span of a set S is the same as the set of all linear combinations of the vectors in S .
- (d) If the set S spans a subspace W of a vector space V , then every vector in W can be written as linear combination of the vectors from S .
- (e) Two vectors are linearly dependent provided that one is a scalar multiple of the other.
- (f) Any subset of a linearly independent set is a linearly dependent set.
- (g) Any set containing the zero vector is linearly independent.
- (h) $\mathbb{R}^n$ has infinitely many vectors, so we say it is infinite-dimensional.
- (i) Two bases for a subspace of $\mathbb{R}^n$ must contain the same number of elements.
- (j) Every set that spans a subspace contains a basis for the subspace.
- (k) The dimension of the nullspace of matrix A is the number of free variables in the general solution for the linear system $A\mathbf{x} = \mathbf{0}$.
- (l) If A is 4×4 with rank $A = 4$, then $A\mathbf{x} = \mathbf{b}$ has exactly four solutions.
- (m) If A and B are row equivalent, then they are similar matrices.
- (n) If a 3×3 matrix A has eigenvalues $\lambda = 1, -1, 3$, then A is diagonalizable.
- (o) If P is invertible and D is diagonal, then the eigenvalues of $A = P^{-1}DP$ are the diagonal entries of D .
- (p) If A is a 3×3 matrix with 2 distinct eigenvalues, then A cannot be diagonalizable.
- (q) For a nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, $\text{proj}_{\mathbf{a}} \mathbf{b}$ is always orthogonal to $\mathbf{a}$.
- (r) If $B = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ is an ordered basis for V , then $C_B(\mathbf{b}_1 + \mathbf{b}_2) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.
- (s) If $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^3$, then $\frac{1}{\|\mathbf{v}\|} \mathbf{v}$ is a unit vector.

2. In $\mathbb{R}^4$, let W be the subset of all vectors $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$ that satisfy $a_4 - a_3 = a_2 - a_1$.

(a) Show W is a subspace of $\mathbb{R}^4$.

(b) Show that $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ spans W .

(c) Find a subset of S that is a basis for W .

(d) Express $\begin{bmatrix} 0 \\ 4 \\ 2 \\ 6 \end{bmatrix}$ as a linear combination of the basis obtained in part (c).

3. Consider

$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 5 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 1 \end{bmatrix}$. Determine whether each vector belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

(a) $\mathbf{v} = \begin{bmatrix} 0 \\ 7 \\ 4 \\ 0 \end{bmatrix}$

(b) $\mathbf{v} = \begin{bmatrix} 5 \\ 6 \\ 2 \\ 1 \end{bmatrix}$

(c) $\mathbf{v} = \begin{bmatrix} 3 \\ -8 \\ -6 \\ 5 \end{bmatrix}$

4. Let A be an $n \times n$ matrix and λ a scalar. Using the definition of subspace, prove that the set W consisting of all vectors $\mathbf{x}$ in $\mathbb{R}^n$ such that $A\mathbf{x} = \lambda\mathbf{x}$ is a subspace of $\mathbb{R}^n$.

5. Consider the subspace of $\mathbb{R}^4$ spanned by

$$S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 4 \\ 1 \end{bmatrix} \right\}$$

- (a) Determine a subset S of the given spanning set that is a basis for W .
- (b) Find a basis T for W that is not a subset of the spanning set.

6. Let V be the set of all 2×1 real matrices with operations

$$\begin{bmatrix} a \\ b \end{bmatrix} \oplus \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ad \\ bc \end{bmatrix}, \quad k \odot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ka \\ kb \end{bmatrix}$$

Is V a vector space? (remember to fully justify)

7. Is $W = \left\{ \begin{bmatrix} a \\ b \\ -a \end{bmatrix} \mid a, b \text{ in } \mathbb{R} \right\}$ a subspace of $\mathbb{R}^3$? (remember to fully justify). If it is, find a basis for W .

8. Let $S = \left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ -6 \end{bmatrix} \right\}$. Determine whether S is a basis for $\mathbb{R}^3$.

9. Find a basis for the nullspace of $A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 2 & 3 & 2 & 1 \\ 1 & 0 & 1 & -1 \end{bmatrix}$

10. If A is 3×3 with $\text{rank } A = 2$, find the dimension of the nullspace of A .

11. Determine all values of c so that the set $\{x + 3, 2x + c^2 + 2\}$ is linearly independent in P_1 .

12. Suppose A is a 3×3 matrix with eigenvalue $\lambda = 2$ and eigenvector $\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$. Compute

$$A^4 \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}.$$

13. For $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$, find a basis for the eigenspace associated with $\lambda = 2$.

14. Find a diagonal matrix similar to $A = \begin{bmatrix} 2 & 1 & -3 \\ 3 & 0 & 3 \\ -1 & 1 & 0 \end{bmatrix}$

15. Is $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ diagonalizable? Explain.

16. Find all vectors in $\mathbb{R}^4$ that are orthogonal to both $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ -2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}$

17. Prove or disprove: The set $W = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| \leq 1\}$ is a subspace of $\mathbb{R}^3$.

18. Let $B = (\mathbf{b}_1, \mathbf{b}_2)$ and $D = (x, x - 1)$ be ordered bases for P_1 . If the transition matrix from B to D is $P_{D \leftarrow B} = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$, determine B .

19. Consider P_3 , and consider the subset $W = \{p(x) \text{ in } P_3 \mid p''(0) - 2p(0) = 0\}$.

(a) Show, using the subspace test, that W is a subspace of P_3 .

(b) Find a basis for W . What is its dimension?